Homeworks

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Specifications-Graded Units

The homework sequence is **7 deliverables** — Unit 0 plus six core units, of which your **best five count** (one drop built in; Unit 0 is always required) — consolidated into connected arcs rather than isolated weekly problem sets. Every deliverable is graded **pass / not-yet** against published specifications: no points, no partial credit, revise-and-resubmit.

Each unit page below gives its objectives, what you'll build, prerequisites, what to submit, and its **rubric** (the same pages are posted in **D2L**). Assignment materials — starter code, autograder, data — are distributed via **Classroom 50**; your submissions live in a private repo visible only to you and the instructor. See Assessment for how the core units set your letter grade: A = any 5, B = 4, C = 3, D = 2.

Unit	Title	You'll build
U0	Tooling & Reproducibility	A working pipeline (clone ⟶ `pytest` ⟶ render ⟶ push) and the seed policy every later unit assumes.
U1	Linear Models & Model Selection	One dataset, three fits: scalar least squares ⟶ matrix normal equations ⟶ cross-validated polynomial.
U2a	Probability & Estimation Foundations (I)	Poisson probabilities + derivations: the Bernoulli MLE and its Fisher-information floor.
U2b	Probability & Estimation Foundations (II)	Written-only: the matrix-calculus identities Unit 1 applies, Gaussian independence, MLE unbiasedness, Beta moments.
U3	Monte Carlo & Estimators	Estimating $\mathbb{E}[f(X)]$ and π by sampling; $O(1/\sqrt{N})$ convergence and the curse of dimensionality.
U4	Bayesian Inference, Exact & Conjugate	Predictive-variance error bars, the exact coin-game posterior, Inverse-Gamma conjugacy.
U5	Approximate Bayesian Inference	One posterior four ways: PGM, MAP (Newton), Laplace, MCMC — and why mode ≠ mean.

How the units connect

The sequence spirals rather than resets: **U0**'s seed policy is graded again in U3 and U5 · **U2b proves** the matrix-calculus identities **U1 applies** · U2b's Beta moments power **U4**'s conjugate priors, and its Hessian identity returns in **U5** · U4's exact posteriors become the targets **U5 approximates** · U4 + U5 together are the method menu the project's Part 2 draws on.